BIRLA INSTITUTE OF TECHNOLOGY

NEP-2020 CURRICULUM BOOK

(Effective from Academic Session: Monsoon 2024)

Bachelor of Technology

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Institute Vision

To become a Globally Recognized Academic Institution in consonance with the social, economic, and ecological environment, striving continuously for excellence in education, research and technological service to the national needs.

Institute Mission

- To educate students at Undergraduate, Post Graduate, Doctoral and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.
- To provide excellent research and development facilities to take up Ph.D. programmes and research projects.
- To develop effective teaching and learning skills and state-of-the-art research potential of the faculty.
- To build national capabilities in technology, education, and research in emerging areas.
- To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

Department Vision

To become an internationally recognized center of excellence in academics, research, and technological services in the area of Electrical and Electronics Engineering and related interdisciplinary fields.

Department Mission

- Imparting strong fundamental concepts to students and motivating them to find innovative solutions to engineering problems independently.
- Developing engineers with managerial attributes capable of applying latest technology with responsibility.
- Creation of congenial atmosphere and excellent research facilities for undertaking quality research by faculty and students.
- To strive for more internationally recognized publication of research papers, books and to obtain patents and copyrights.
- To provide excellent technological services to industry.

Programme Educational Objectives (PEOs)

- To develop capability to understand the fundamentals of Science and Electrical & Electronics Engineering for analyzing engineering problems with futuristic approach.
- To foster a confident and competent graduate capable of solving real life practical engineering problems fulfilling the obligation towards society.
- To inculcate an attitude for identifying and undertaking developmental work both in industry as well as in academic environment with emphasis on continuous learning enabling to excel in competitive participations at global level.
- To nurture and nourish effective communication and interpersonal skills to work in a team with a sense of ethics and moral responsibility for achieving goal.

Programme Outcomes (POs)

- Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
- Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
- Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
- The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
- Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- Life-Long Learning: Recognize the need for and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies and critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

- Pragmatic professional exposure in the domain of Power Engineering, Power Electronics, Measurement, and Control through systematically designed courses, laboratory contents, projects, and commensurate electives.
- Development of analytical and designed skills for electrical & electronic systems as well as multidisciplinary areas of engineering utilizing appropriate conventional and modern tools.
- Holistic amalgamation of managerial skills and research aptitude to strengthen industry-academia collaboration.

SEMESTER 1 - FIRST

Total Credits: 22

Course Code	Subject	L	T	P	Credits
MA24101	Mathematics - I (FS)	3	1	0	4
CH24101	Chemistry (FS)	3	1	0	4
EC24101	Basic Electronics (GE)	2	1	0	3
ME24101	Basics of Mechanical Engineering (GE)	2	1	0	3
CE24101	Environmental Sciences (FS)	2	0	0	2
CH24102	Chemistry Lab (FS)	0	0	2	1
EC24102	Basic Electronics Lab (GE)	0	0	2	1
ME24102	Engineering Graphics (GE)	0	0	4	2
PE24102	Workshop Practice (GE)	0	0	2	1
MC24101-105	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship (MC)	0	0	2	1

SEMESTER 2 - SECOND

Total Credits: 22.5

Course Code	Subject	L	T	P	Credits
MA24103	Mathematics - II (FS)	3	1	0	4
PH24101	Physics (FS)	3	1	0	4
BE24101	Biological Sciences for Engineers (FS)	2	0	0	2
CS24101	Programming for Problem Solving (GE)	3	1	0	4
EE24101	Basics of Electrical Engineering (GE)	2	1	0	3
PH24102	Physics Lab (FS)	0	0	2	1
CS24102	Programming for Problem Solving Lab (GE)	0	0	2	1
EE24102	Electrical Engineering Lab (GE)	0	0	2	1
HS24131	Communication Skill - I (HSS)	0	0	3	1.5
MC24106-110	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship (MC)	0	0	2	1

Grand Total for First Year: 44.5 Credits

Vocational Courses for Exit after 1st Year

Course Code	Course Name	L	T	P	Credits
EE24151	Basics of Electrical Appliance and Electrical Wiring	3	1	0	4
EE24152	Laboratory on Electrical System Installation, Maintenance and Troubleshooting	0	0	3	2

SEMESTER 3 - THIRD

Total Credits: 23

Course Code	Subject	L	T	P	Credits
EE24201	Electrical Measurement and Instrumentation (PC)	3	0	0	3
EE24203	Analog and Digital Circuits (PC)	3	0	0	3
EE24205	Engineering Electromagnetics (PC)	3	1	0	4
EE24207	Introduction to System Theory (PC)	2	1	0	3
EE24209	Circuit Theory (PC)	3	1	0	4
MT24131	UHV-II: Understanding Harmony (HSS)	3	0	0	3
EE24202	Electrical Measurement and Instrumentation Lab (PC)	0	0	2	1
EE24204	Analog and Digital Circuits Laboratory (PC)	0	0	2	1
MC24201-205	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship (MC)	0	0	2	1

SEMESTER 4 - FOURTH

Total Credits: 24.5

Course Code	Subject	L	T	P	Credits
EE24251	DC Machines and Transformer (PC)	3	1	0	4
EE24253	Digital Signal Processing (PC)	3	1	0	4
EE24255	Microprocessor and Embedded System (PC)	3	0	0	3
EE24257	Electrical Energy Generation and Control (PC)	3	0	0	3
OE/MOOC	Open Elective - I / MOOC - I (OE)	3	0	0	3
MA24201	Numerical Methods (GS)	2	0	0	2
HS24211	Indian Knowledge System (MC)	2	0	0	0
EE24252	Electrical Machine Lab - I (PC)	0	0	3	1.5
EE24254	Digital Signal Processing Laboratory (PC)	0	0	2	1
EE24256	Microprocessor and Embedded System Lab (PC)	0	0	3	1
MA24202	Numerical Methods Laboratory (GS)	0	0	2	1
MC24206-210	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship (MC)	0	0	2	1

Grand Total for Second Year: 47.5 Credits

Vocational Courses for Exit after 2nd Year

Course Code	Course Name	L	T	P	Credits
EE24275	Electrical Power Equipment and Protection	3	1	0	4
EE24276	Laboratory on Electrical Power Equipment and Protection	0	0	3	2

SEMESTER 5 - FIFTH

Total Credits: 24

Course Code	Subject	L	T	P	Credits
EE24301	AC Rotating Machines (PC)	3	0	0	3
EE24303	Control Theory (PC)	3	1	0	4
EE24305	Power Electronics (PC)	3	1	0	4
EE24307	Electric Power Transmission and Distribution (PC)	3	0	0	3
PE	Program Elective - I (PE)	3	0	0	3
OE/MOOC	Open Elective - II / MOOC - II (OE)	3	0	0	3
EE24302	Electrical Machine Lab (PC)	0	0	2	1
EE24304	Control System Laboratory (PC)	0	0	2	1
EE24300	Project - I (PC)				2

SEMESTER 6 - SIXTH

Total Credits: 22.5

Course Code	Subject	L	T	P	Credits
EE24351	Power System Analysis (PC)	3	0	0	3
EE24353	Electric Drives (PC)	3	0	0	3
PE	Program Elective - II (PE)	3	0	0	3
PE	Program Elective - III (PE)	3	0	0	3
OE/MOOC	Open Elective - III / MOOC - III (OE)	3	0	0	3
MT24204	Constitution of India (HSS)	2	0	0	0
EE24306	Power Electronics Laboratory (PC)	0	0	2	1.5
EE24356	Electrical Workshop (PC)	0	0	2	1
PE Lab	Program Elective - III Lab (PE)	0	0	3	1.5
EE24350	Project - II (PC)				2
HS24133	Communication Skill - II (HSS)	0	0	3	1.5

Grand Total for Third Year: 46.5 Credits

SEMESTER 7 - SEVENTH

Total Credits: 22.5

Course Code	Subject	L	T	P	Credits
EE24401	Switchgear and Protection (PC)	3	1	0	4
PE	Program Elective - IV (PE)	3	0	0	3
PE	Program Elective - V (PE)	3	0	0	3
OE/MOOC	Open Elective - IV / MOOC - IV (OE)	3	0	0	3
EE24402	Power System Laboratory (PC)	0	0	2	1
PE Lab	Program Elective - V Lab (PE)	0	0	3	1.5
MC24400	Summer Training (Min 4 Weeks/160 Hrs) (MC)				4
EE24400	Project - III (PC)				3

SEMESTER 8 - EIGHTH

Total Credits: 7

Course Code	Subject	L	T	P	Credits
EE24450/EE24490	Project - IV / Industry Internship (PC)				6
EE24498	Comprehensive Viva (PC)				1

Grand Total for Fourth Year: 29.5 Credits

GRAND TOTAL FOR B.TECH: 168 Credits

LIST OF PROGRAM ELECTIVES (PE) COURSES

Programme Elective - I (Semester 5)

- EE24325 - Sensors and Transducers
- EE24327 - Fundamentals of Communication System
- EE24328 - Electrical Engineering Materials
- EE24329 - Computational Techniques in Electrical Engineering
- EE24331 - Embedded Systems and Applications
- EE24333 - Industrial Instrumentation

Programme Elective - II (Semester 6)

- EE24371 - EHV AC Power Transmission
- EE24373 - Sustainable Energy Sources
- EE24375 - Sensing Technology and Applications
- EE24377 - Power Semiconductor Devices
- EE24381 - Internet of Things
- EE24383 - Machine Learning
- EE24385 - Artificial Intelligence for Electrical Engineering
- EE24387 - Special Electrical Machines

Programme Elective - III (Semester 6)

- EE24385 - Bioinstrumentation and Concepts
- EE24387 - Applied Control Theory
- EE24386 Lab - Bioinstrumentation Lab
- EE24388 Lab - Applied Control Lab

Programme Elective - IV (Semester 7)

- EE24411 - Specifications & Estimation of Electrical Installations
- EE24421 - Robotics
- EE24413 - Utilization of Electrical Power
- EE24415 - High Voltage Engineering
- EE24417 - HVDC and FACTS
- EE24419 - Testing & Commissioning of Electrical Equipment
- EE24XXX - Microgrid Operation and Control
- EE24423 - Advanced Power System Analysis and Control
- EE24425 - Nonlinear and Adaptive Control
- EE24427 - Power Conversion Techniques
- EE24459 - Smart Power System
- EE24429 - Power Electronics Application
- EE24431 - Electric Vehicles
- EE24433 - Blockchain Application in Power Systems
- EE24437 - Embedded Control of Switching Power Converters
- EE24439 - Introduction to Autonomous Vehicles
- EE24441 - Introduction to Demand Side Management

Programme Elective - V (Semester 7)

- EE24451 - Electrical Machine Design
- EE24457 - Hybrid Electric Vehicle
- EE24453 - Computer Aided Power System Analysis
- EE24455 - Advanced Power Electronics
- EE24452 Lab - Computer Aided Electrical Machine Design Lab
- EE24454 Lab - Computer Aided Power System Analysis Lab
- EE24456 Lab - Advanced Power Electronics Lab
- EE24458 Lab - Electric Vehicle Lab
- EE24460 Lab - Smart Grid Lab